Best Practices for Biomedical Research Data Management

Module 7: Data Sharing and Reuse
Impact of Research Data

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This video will discuss many methods used to track research impact, including traditional citation-based metrics, article level or alternative metrics, and measuring the “digital footprint” of research (datasets, software, blogs , etc.). The video will also highlight tools and services for tracking research impact metrics.

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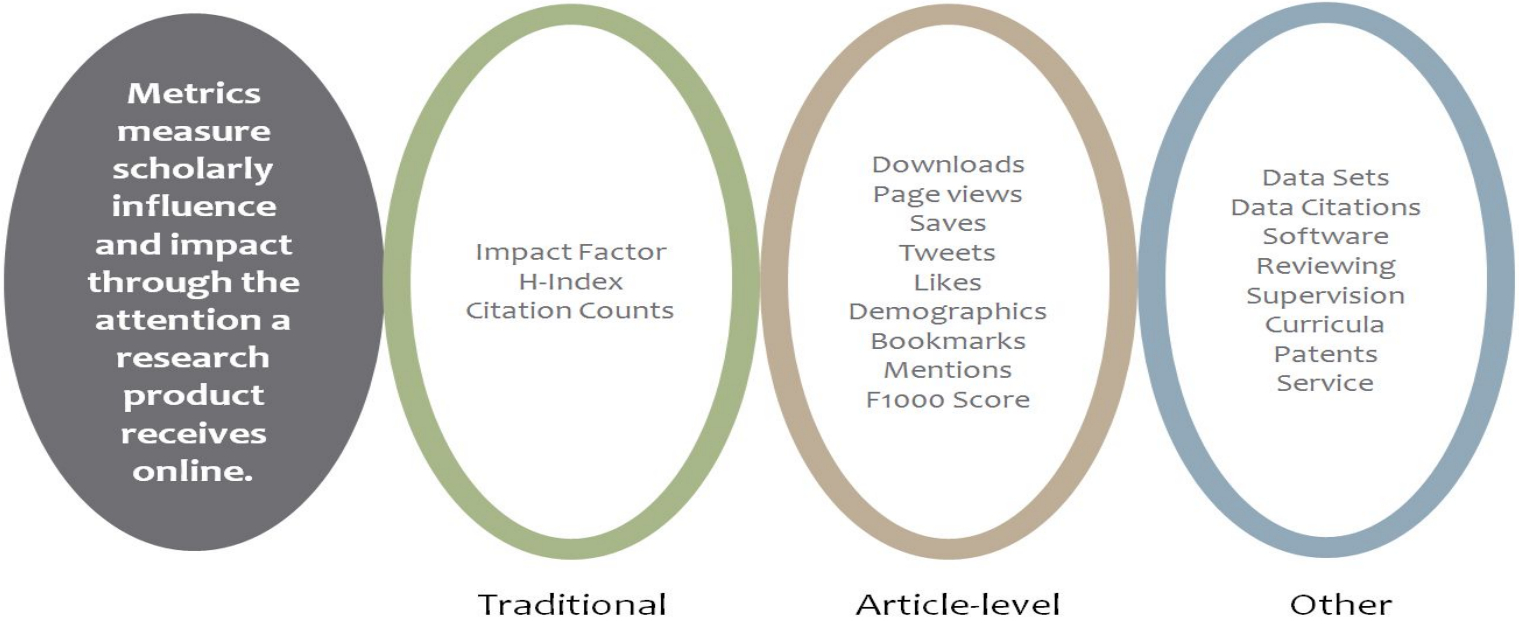
Tracking research impact



Why measure impact of data?

To measure our influence in our communities

To evaluate the performance of ourselves, our departments, our peers

To have tangible evidence of benefit to weigh against the costs of research

To incentivize data sharing

To provide researchers with a more comprehensive understanding of their readership, collaboration opportunities, public engagement

Metrics for data

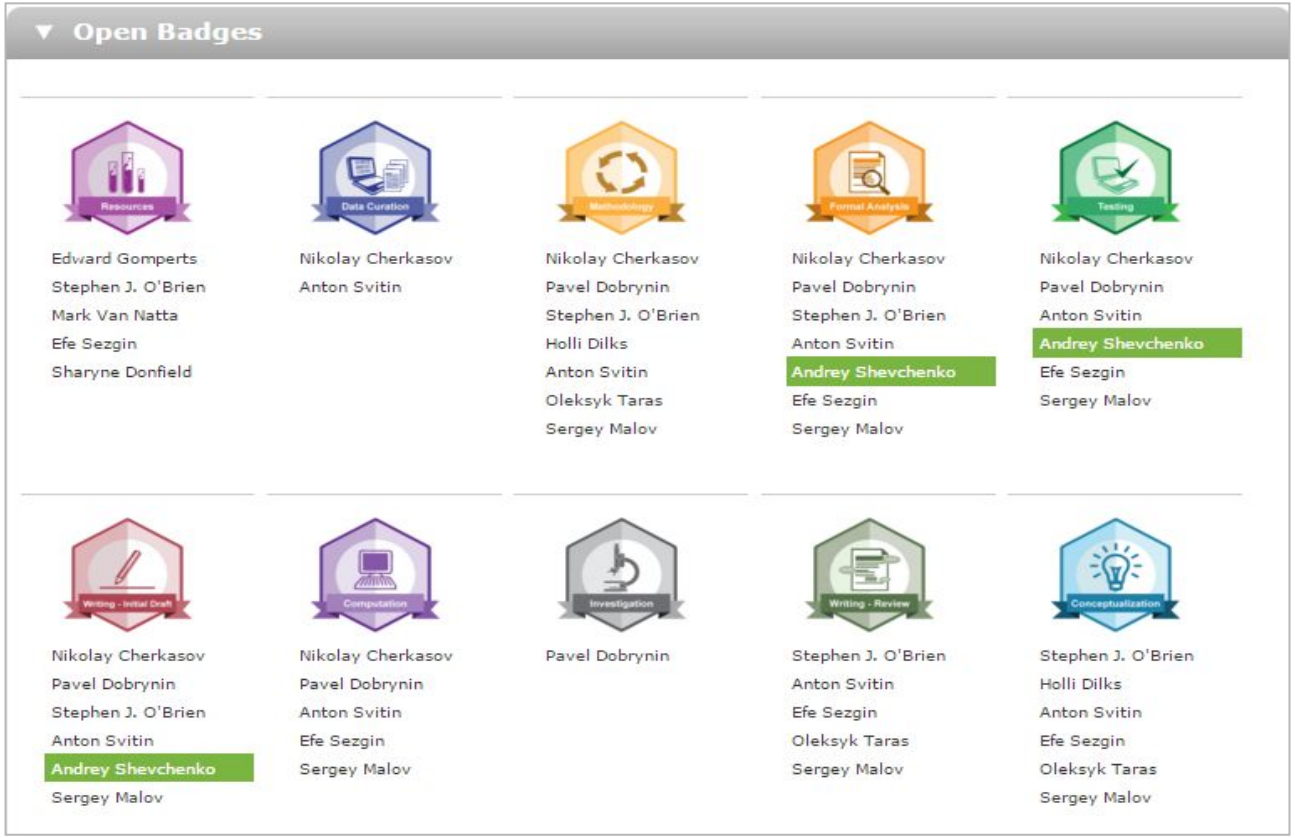
- Citations

- Resolutions
- Page views
- Downloads
- Social media links
- Digital badges

Alternative metrics (altmetrics, article level metrics, webometrics, cybermetrics): measurements of attention and dissemination that go beyond (and are complementary to) traditional bibliometrics, often focusing on social media tools

Becker Medical Library Model for Assessment of Research Impact. https://becker.wustl.edu/impact-assessment
Alex Ball and Monica Duke, Digital Curation Centre (2015). How to track the impact of research data with metrics. http://www.dcc.ac.uk/resources/how-guides/track-data-impact-metrics

Author contributorship badges



GigaBlog (28 September 2015). Putting credit back into the hands of researchers. http://blogs.biomedcentral.com/gigablog/2015/09/28/putting-credit-hands-researchers/

Nature (12 May 2016). Digital badges motivate scientists to share data. http://dx.doi.org/10.1038/nature.2016.19907.

Tools and services

Altmetric.com

Altmetric

? What is this page? | Embed badge | Share

Dreadnoughtus schrani 3D PDF images - Lacovara et al., 2014, A Gigantic, Exceptionally Complete Titanosaurian Sauropod Dinosaur from Southern Patagonia, Argentina, Scientific Reports,

Overview of attention for research output published on figshare, September 2014

185

About this Attention Score

In the top 5% of all research outputs scored by Altmetric

MORE...

Mentioned by

- **4** news outlets
- **2** blogs
- **166** tweeters
- **2** Facebook pages
- **5** Google+ users
- **1** video uploader

SUMMARY | News | Blogs | Twitter | Facebook | Google+ | Video

Title	Dreadnoughtus schrani 3D PDF images - Lacovara et al., 2014, A Gigantic, Exceptionally Complete Titanosaurian Sauropod Dinosaur from Southern Patagonia, Argentina, Scientific Reports,
Published on	figshare, September 2014
DOI	10.6084/m9.figshare.1130885
Authors	Kenneth Lacovara, Kenneth, Lacovara

View on publisher site

Alert me about new mentions

TWITTER DEMOGRAPHICS | ATTENTION SCORE IN CONTEXT

This research output has an **Altmetric Attention Score** of **185**. This is our high-level measure of the quality and quantity of online attention that it has received. This Attention Score was calculated when the research output was last mentioned on **29 June 2016**.

ALL RESEARCH OUTPUTS	OUTPUTS FROM FIGSHARE	OUTPUTS OF SIMILAR AGE	OUTPUTS OF SIMILAR AGE FROM FIGSHARE
#20,497	#11	#898	#2
of 5,462,231 outputs	of 8,505 outputs	of 140,106 outputs	of 487 outputs

Altmetric has tracked 5,462,231 research outputs across all sources so far. Compared to these this one has done particularly well and is in the 99th percentile: it's **in the top 5% of all research outputs ever tracked** by Altmetric.

http://www.altmetric.com/details/2653335#score

ImpactStory profile

Carl Boettiger

Theoretical Ecologist. Bio at: http://carlboettiger.info

Open Access

UC Berkeley

Overview
Map
Twitter fans

articles (15)
datasets (16)
figures (1)
posters (3)
slide decks (17)
software products (77)
theses (4)

Selected works

Tipping points: From patterns to predictions.
(2013) Boettiger, Hastings. *Nature*
read fulltext
highly cited | highly saved | highly discussed | highly viewed

Lab Notebook, 2012
(2013) *figshare.*
highly viewed +11 | highly discussed | highly viewed

Key profile metrics

1.6k saves across 15 articles

148 forks across 77 software products

2.8k views across 15 articles

385 tweets across 15 articles

http://v1.impactstory.org/CarlBoettiger

ImpactStory dataset metrics

http://v1.impactstory.org/CarlBoettiger/product/0omfaijgw5b6gcs4os6tk70g/metrics

Recommendations for increasing impact

Identify yourself (register for an ORCID)

Deposit data in a trustworthy repository

Provide appropriate metadata

Enable open access

Apply a license to the data

Raise awareness

Free ebook

https://becker.wustl.edu/impact-assessment/strategies